

复变函数笔记

只有后一部分笔记

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序

主要是从复积分以后的笔记,第三章前也需要好好复习,特别是复微分和级数,可能有错,建议使用 ai 辅助.

$$z^{\frac{1}{n}} = \sqrt[n]{r} e^{i \frac{\theta + 2k\pi}{n}} \quad (k = 0, 1, 2, \dots, n)$$

指数和对数转换

$$\log z = \ln(|z|) + i(\operatorname{Arg}(z) + 2k\pi)$$

$$\operatorname{Log} z = \ln(|z|) + i \operatorname{Arg}(z)$$

体现多值性

$$\log e^z = z + i(2k\pi)$$

体现周期性

$$e^{\log z} = z$$

例题

计算 $(1+i)^{1-i}$

$$(1+i)^{1-i} = e^{\log(1+i) \times (1-i)} = e^{(\ln \sqrt{2} + i(\frac{\pi}{4} + 2k\pi)) \times (1-i)}$$

$$= e^{\ln \sqrt{2} + \frac{\pi}{4} + 2k\pi + i(\frac{\pi}{4} + 2k\pi - \ln \sqrt{2})}$$

$$= \sqrt{2} e^{\frac{\pi}{4} + 2k\pi} \left[\cos\left(\frac{\pi}{4} - \ln \sqrt{2}\right) + i \sin\left(\frac{\pi}{4} - \ln \sqrt{2}\right) \right] \quad (k = 0, \pm 1, \pm 2 \dots)$$

解析和可导

- 可导:只要求某一个点处的导数存在,证明可导:满足 CR 且 u, v 在 (x, y) 可微
- 解析:在某个邻域内处处可导:满足 CR 且 u, v 在 D 可微

结论:化简到最后含有 z 的共轭一定不解析

柯西黎曼定理

Note

Suppose that

$$f(z) = u(x, y) + iv(x, y)$$

and that $f'(z)$ exists at a point $z_0 = x_0 + iy_0$. Then the first-order partial derivatives of u and v must exist at (x_0, y_0) , and they must satisfy the **Cauchy-Riemann equations**

$$u_x = v_y \quad u_y = -v_x$$

, Also, $f'(z_0)$ can be written $f'(z_0) = u_x + iv_x$, where these partial derivatives are to be evaluated at (x_0, y_0)

必要条件, 如果可导, 那么偏导存在且满足柯西黎曼方程, 反之不一定成立.

柯西古萨定理

单个联通区域

Note

Let α be a simple contour. Suppose that $f(z)$ is analytic in the interior of α and on α . Then

$$\int_{\alpha} f(z) dz = 0$$

多个联通区域

Note

Suppose that

- (a) C is a simple closed contour described in the counterclockwise direction
- (b) $C_k (k = 1, 2, \dots, n)$ are simple closed contours interior to C , all described in the clockwise direction, that are disjoint and whose interior have no points in common

If a function f is analytic on all of these contours and throughout the multiply connected domain consisting of the points inside C and exterior to each C_k , then

$$\int_C f(z) dz + \sum_{k=1}^n \int_{C_k} f(z) dz = 0$$

柯西积分公式

Let f be analytic inside and on a simple closed contour C , taken in the positive sense. If z_0 is any point interior to C , then

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)dz}{z - z_0}$$

Let f be analytic inside and on a simple closed contour C , taken in the positive sense. If z_0 is any point interior to C , then

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)dz}{(z - z_0)^{n+1}}$$

级数

taylor 级数

Note

Suppose that a function f is analytic throughout a disk $|z - z_0| < R_0$, centered at z_0 and with radius R_0 . Then $f(z)$ has the power series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad (|z - z_0| < R_0)$$

where

$$a_n = \frac{f^{(n)}(z_0)}{n!} \quad (n = 0, 1, 2, 3, \dots)$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad (|z| < 1)$$

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad (|z| < \infty)$$

$$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad (|z| < \infty) \quad \cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \quad (|z| < \infty)$$

$$\sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} \quad (|z| < \infty) \quad \cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} \quad (|z| < \infty)$$

laurent 级数

Note

Suppose that a function f is analytic throughout an annular domain $R_1 < |z - z_0| < R_2$ centered at z_0 , and let C denote any positive oriented simple closed contour around z_0 and lying in that domain. Then, at each point in the domain, $f(z)$ has the series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \quad (R_1 < |z - z_0| < R_2)$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z) dz}{(z - z_0)^{n+1}} \quad (n = 0, 1, 2, \dots)$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(z) dz}{(z - z_0)^{-n+1}} \quad (n = 1, 2, \dots)$$

乘除法

例题

求 $\tan z = \frac{\sin(z)}{\cos(z)}$ 的留数

$$\cos z = 1 - \frac{z^2}{2} + \frac{z^4}{24} + \dots$$

$$\sin z = z - \frac{z^3}{6} + \frac{z^5}{120} + \dots$$

假设

$$\tan z = \sum_{k=0}^{\infty} a_k z^k$$

$$z - \frac{z^3}{6} + \frac{z^5}{120} + \dots = \sum_{k=0}^{\infty} a_k z^k \times \left(1 - \frac{z^2}{2} + \frac{z^4}{24} + \dots \right)$$

比较系数可得 $a_0 = 0, a_1 = 1, a_2 = 0, a_3 = \frac{1}{3}, a_4 = 0, a_5 = \frac{2}{15}$

留数

奇点和零点

孤立奇点:在其邻域内没有其他奇点

$f(z) = \frac{1}{\sin(\frac{\pi}{z})}$ $z = 0$ 不是孤立奇点,因为 $z = \frac{1}{k} \rightarrow 0$ 所以0周围邻域有奇点.

三种孤立奇点:

1. 可去奇点

洛朗展开的时候所有 b_i 都是 0

2. 本质奇点

洛朗展开的时候有无限的 b_i ,本质奇点的倒数也是本质奇点

3. m 阶极点

洛朗展开的时候有限个 m 个 b_i

例题

判断函数 $\frac{z-1}{(z^2+1)z^2 \sin(\pi z)}$ 奇点的类型

候选奇点为 $z = \pm i, z = 0, z = k \quad (k = 0, 1, 2, 3...), z = \infty$

1. $z = \pm i$ 1 阶极点
2. $z = 0$ 是 3 阶极点,不同项产生的要合并
3. $z = k \quad (k = 1, 2, 3...)$ 1 阶极点
4. $z = \infty \quad f(\frac{1}{z}) = \frac{\frac{1}{z}-1}{(\frac{1}{z^2}+1)\frac{1}{z^2} \sin(\pi \frac{1}{z})}$ 本质奇点,因为 $\sin(\pi \frac{1}{z})$ 是本质奇点

求本质奇点的留数

只有一种方式,展开到洛朗级数,然后看 b_{-1}

例题

求下面函数在 0 处的留数

$$f(z) = e^{\frac{1}{z}}$$

$$f(z) = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \dots$$

由此可知

$$\operatorname{Res}(f(z), 0) = 1$$

求极点的留数

定义方法

求洛朗展开后 b_{-1} 之前的系数

极限方法

$$f(z) = \frac{\varphi(z)}{(z-z_0)^m}$$

$$\operatorname{Res}(f(z), z_0) = \begin{cases} \lim_{z \rightarrow z_0} \varphi(z) & m = 1 \\ \lim_{z \rightarrow z_0} \frac{\varphi(z)^{(m-1)}}{(m-1)!} & m \geq 2 \end{cases}$$

零点方法

如果是 $f(z) = \frac{p(z)}{q(z)}$ 的形式, p, q 为多项式, $q(z_0) = 0, q'(z_0) \neq 0, p(z_0) \neq 0$,可以考虑

$$\operatorname{Res}(f(z), z_0) = \frac{p(z_0)}{q'(z_0)}$$

无穷远处留数

性质 1:

$$\sum_{k=1}^n \operatorname{Res}_{z=z_k} (f(z), z) + \operatorname{Res}(f(z), \infty) = 0$$

性质 2:

$$\operatorname{Res}(f(z), \infty) = - \operatorname{Res}\left(f\left(\frac{1}{z}\right) \times \frac{1}{z^2}, 0\right)$$

例题

计算积分 $I = \int_C \frac{z^{10}}{(z^{10}-2i)(z-3i)} dz$, 其中 $C: |z| = 2$, 正向曲线

奇点为 $z_0 = 3i, z_1, \dots, z_{10} = (2i)^{\frac{1}{10}}$ 发现 z_0 不在 C 内部.

无穷处的留数 + z_0 处留数 + z_1 处留数 + $\dots$ + z_{10} 处留数 = 0

只需要考虑无穷处的留数和 z_0 处的留数, 从而算出其他 10 个留数的和

$$\operatorname{Res}\left(\frac{z^{10}}{(z^{10}-2i)(z-3i)}, \infty\right) = - \operatorname{Res}\left(\frac{1}{(1-3iz)(1-2iz^{10})z}, 0\right) = -(-1) = 1$$

$$\operatorname{Res}\left(\frac{z^{10}}{(z^{10}-2i)(z-3i)}, z_0\right) = -\frac{3^{10}}{-3^{10}-2i}$$

所以结果为 $I = (-B_1 - B_2) \times 2\pi i = \left(1 - \frac{3^{10}}{-3^{10}-2i}\right) 2\pi i$

柯西留数定理

Note

Let C be a simple closed contour, described in the positive sense. If a function f is analytic inside and on C except for a finite number of singular points z_k ($k = 1, 2, \dots, n$) inside C , then

$$\int_C f(z) = \sum_{k=1}^n 2\pi i \operatorname{Res}_{z=z_k} f(z)$$

例题

$$\int_{|z|=3} \frac{z^3}{(z+1)(z+4)} e^{\frac{1}{z}} =$$

1. 计算奇点 $z = -1, z = -4, z = 0$
2. 看在不在积分区域内, 只有 $z = -1$ 和 $z = 0$
3. 判断奇点的类型, $z = -1$ 是 1 阶极点, $z = 0$ 是本质奇点, 没有可去奇点
4. 对于可去奇点, 忽略掉
5. 对于本质奇点, 做 laurent 展开, 求 b_{-1}
6. 对于极点, 使用定义法, 极限法或零点法求留数
7. 使用柯西留数定理求积分结果

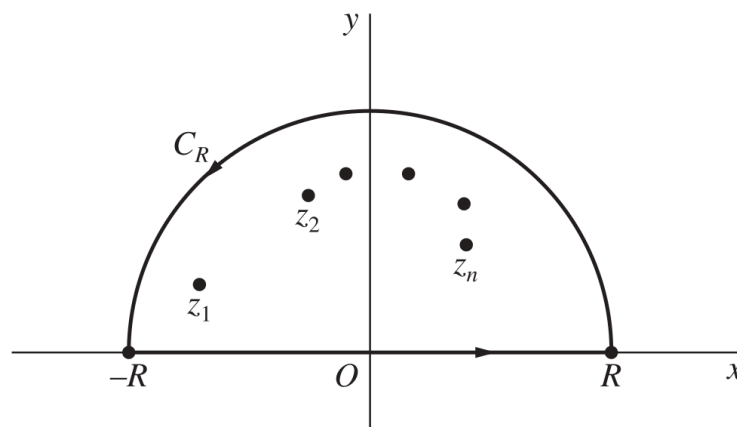
留数应用

$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx$ 类瑕积分计算

计算实数瑕积分

$$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx$$

其中 $\deg(q(x)) \geq \deg(p(x)) + 2$



有

$$\int_{-R}^R \frac{p(x)}{q(x)} dx + \int_{C_R} \frac{p(x)}{q(x)} dx = 2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k}(f(z))$$

所以

$$\int_{-R}^R \frac{p(x)}{q(x)} dx = 2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k}(f(z)) - \int_{C_R} \frac{p(x)}{q(x)} dx$$

$$\int_{C_R} \frac{p(x)}{q(x)} dx = \pi R \left| \frac{p(x)}{q(x)} \right| = O\left(\pi R \frac{1}{R^2}\right)$$

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{p(x)}{q(x)} dx = 0$$

所以

$$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx = 2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k}(f(z))$$

当函数为偶函数,

$$\operatorname{PV} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{p(x)}{q(x)} dx = \pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k}(f(z))$$

做题的时候注意要选取在上半平面的留数计算.

$\int_{-\infty}^{\infty} f(x) \cos(ax) dx$ 类瑕积分计算

$\int_{-\infty}^{\infty} f(x) \cos(ax) dx$ 和 $\int_{-\infty}^{\infty} f(x) \sin(ax) dx$ 的计算在于构造

$$\int_{-\infty}^{\infty} f(x) e^{iax} dx$$

因为 $\Re(e^{iax}) = \cos(ax)$, $\Im(e^{iax}) = \sin(ax)$ 所以只需要最后使得两侧 $\Re$ 或 $\Im$ 相等即可.

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \cos(ax) dx &= \Re \left(\int_{-\infty}^{\infty} f(z) e^{iaz} dz \right) \\ &= \Re \left(2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k} (f(z) e^{iaz}) - \int_{C_R} f(z) e^{iaz} dz \right) \end{aligned}$$

需要使得 $\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{iaz} dz = 0$

$$\Re \left(\int_{C_R} f(z) e^{iaz} dz \right) \leq \pi R |e^{iaz}| \leq \pi R e^{-ay}$$

推出

$$\Re \left(\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{iaz} dz \right) = 0$$

所以

$$\int_{-\infty}^{\infty} f(x) \cos(ax) dx = \Re \left(2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k} (f(z)e^{iaz}) \right)$$

$$\int_{-\infty}^{\infty} f(x) \sin(ax) dx = \Im \left(2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k} (f(z)e^{iaz}) \right)$$

做题的时候注意要选取在上半平面的留数计算.

Jordan' s Lemma

1. $a > 0, C_R$ 是上半圆弧
2. 在上半圆 C_R 上, $f(z)$ 一致有界: $|f(z)| \leq M$

那么可以得到

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{iaz} dz = 0$$

可以理解为, 指数因子 e^{iaz} 在上半圆上衰减, 使得弧线积分消失

所以证明弧线积分为 0, 只需要证明 $f(z)$ 有界, 即找到 M , 使得 $|f(z)| \leq M$.

含三角函数积分的计算

对于

$$\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$$

$$\text{令 } z = e^{i\theta}, d\theta = \frac{dz}{iz}, \cos \theta = \frac{z+z^{-1}}{2}, \sin \theta = \frac{z-z^{-1}}{2}$$

带入转化为求

$$\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta = \int_{|z|=1} R\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2}\right) \frac{dz}{iz}$$

结论

$$\int_0^{2\pi} \frac{1}{a + b \cos \theta} d\theta = \frac{2\pi \operatorname{sgn}(a)}{\sqrt{a^2 - b^2}}$$

$$\int_0^{2\pi} \frac{1}{a + a \sin \theta + b \cos \theta} d\theta = \frac{2\pi \operatorname{sgn}(a)}{\sqrt{a^2 - (b^2 + c^2)}}$$